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IoT-Based Fall Detection System for the Elderly Using Machine Learning Algorithms

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***Abstract*—Those who are elderly face a significant danger to their health when they fall. On the other hand, if the situation is not brought to light promptly, it can result in the loss of life or disability of senior people, which in turn lowers their quality of life. In this research, we propose a solution to this criteria by presenting a Fall Detection System that is based on the Internet of Things (FallDS-IoT). These systems are designed to detect falls that occur among senior individuals through the use of a wearable system. To obtain precise results while detecting falls, we make use of accelerometers and gyroscopes with our sensors. We divide the actions that elderly people engage in daily into 4 categories: walking, sitting, sleeping, and falling. Specifically, the K-Nearest Neighbours algorithm and the decision-tree algorithm are the 2 recognizable ML algorithms that we employ to handle the work described above. Accuracy rates of 97.85 percent and 89.65 percent, respectively, were gained by the dataset. In light of this, we were able to conclude that K-NN provides a higher level of accuracy when detecting falls, and that this method applies to categorization. Every time a fall happens, a Python script will send a preset message to a registered phone number informing about the incident. This message will be delivered whenever a fall occurs.**

***Keywords*—Fall-DS; Healthcare; Machine Learning; IoT; Data Analytics.**

I. Introduction

The majority of injuries and fatalities that occur are the result of falls. Several factors, including physical disabilities, heart attacks, low blood pressure, and other conditions, might cause older adults living at home to experience falls. In light of this, the senior fall rate likewise increases as the age of the population grows. With the advent of smartphones, senior citizens can call or text someone in the event of an emergency [1]. If, on the other hand, the circumstance is even more dire, they will be unable to use their smartphone. Smartphones are a relatively new technology, but the majority of senior people do not use them. As a result, there ought to be an automatic fall detection system that is capable of precisely detecting the fall and that would send an automated message in the event of an emergency [2]. To take care of fall incidents, it is necessary to have an automated system installed in the home [3]. This system should be able to send a message as promptly as possible in the event of an emergency. To achieve this objective, a Fall DS that is wearable and utilizes the Internet of Things (FallDS-IoT) has been developed. This system is capable of properly detecting fall circumstances within the home [4], [5].

In the Internet of Things (IoT), a compilation of devices is used to construct a valid network for specific applications. Specifically, an accelerometer, a gyroscope, and an Arduino Uno are the low-power sensors that are utilized by the fall-DS system, which is a wearable device that is based on the IoTs and is designed to discover falls in senior individuals [6], [7]. A gyroscope and a three-dimensional accelerometer are coupled to a wearable gadget that is specifically designed to gather data from the movements of older persons using Python if a fall occurs.

A person's life could be saved in the event of a fall, which is the primary reason for the implementation of the fall DS. Numerous studies have been presented in this field that make use of the Internet of Things (IoT) and ML. With that being said, the development of a FallDS-IoT which is low-cost for a smart-house will be advantageous for those who are in need [8], [9]. FallDS-it is a well-established wearable device that is capable of capturing even the tiniest amount of falls. Because of this, individuals can react as rapidly as they possibly can. It is the primary objective of FallDS-IoT to generate a warning before the occurrence of significant harm [10], [11]. The Internet of Things (IoT) offers a potential

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solution for a fall-DS that makes use of sensors [12]. Python programming, and the Arduino platform to accurately identify falls by employing familiar ML algorithms. In addition, we make utilize of classification methods to conduct additional analysis and prediction. We test the proposed system by utilizing a dataset that was developed by us. It is important to take note that the dataset that was generated has 20,500 instances, which includes all of the measurement data. The decision tree algorithms and K-nearest neighbors are the ones that we have decided to utilize for analysis because they are more effective when applied to smaller datasets [13].

This work makes the following contributions, which are stated as follows: The purpose of a wearable FallDS-IoT is to make sure a before warning of the occurrence of a fall, to allow individuals to be taken. Then, the MPU6050 sensor, which is equipped with both a gyroscope and an accelerometer, is the sensor that was utilized in the construction of the wearable FallDS-IoT [14], [15]. These conditions are referred to as sleeping, sitting, walking, and falling. For training, a dataset size of 20,500 instances was utilized. In addition, the dataset that was generated is utilized for training purposes, with a decision tree, and K-nearest neighbors being utilized to classify the incoming inputs that are received from the sensor. The K-NN and decision tree models can categorize the latest inputs with an accuracy of 97.85 percent and 91.39 percent, respectively, respectively, during the testing phase [16]. Because decision trees are the most accurate method for determining the actual accuracy of K-NN, K-NN is also used for the classifications [17].

An explanation of the remaining parts of the paper is provided below. The paper's approach is discussed in Section II of the document. In the third section, a flowchart is used to describe the illustration of the FallDS-IoT system. In the fourth section, the results are presented. In conclusion, we will get to Section V at the end .

II. Method

In this part of study, we will briefly cover the 2 ML techniques that are used for IoTs datasets, namely decision trees, and K-NN, as well as the dataset that is utilized [26].

A. Machine Learning Techniques

Utilizing techniques that are associated with machine learning (data analytics) has as its primary goal the extraction of features from the data that is created. An algorithm known as the K-Nearest Neighbours (K-NN) [18]: It is a well-known supervised learning technique that is used for the study of the Internet of Things (IoT). In this context, the letter K represents the total data points that are specifically selected. It is also known as a lazy learner since it does not undergo any phase of specialized instruction [19]. The distance method is the most important component of the K-NN algorithm. Its purpose is to determine the distance that exists among the datasets. For instance, the Euclidean, Manhattan, or Minkowski distance approaches may be taken into consideration when doing an analysis [20]. It concentrates its attention largely that are

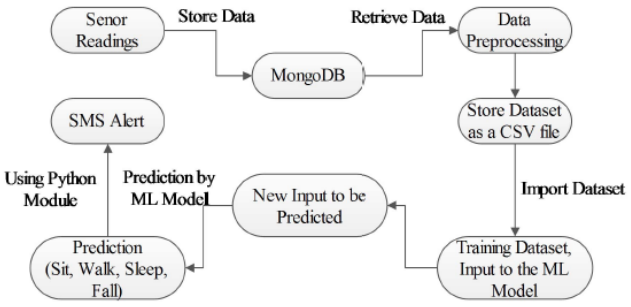


Fig. 1. Schematic Representation of the FallDS-IoT System

the closest to the total collection of points. The data point, unknown, is then merged into a certain class, with the majority of the K data points serving as the basis for the assignment.

This technique is known as the decision tree. The decisions are represented in the view of a flowchart or tree structure, depending on the style of the structure. In the field of operations research, it is particularly utilized for decision-making purposes. A collection of decision rules serves as the foundation for it. Both the Gini index and information gain are utilized in the decision tree to divide the sample into two or more sets that are similar to one another [21]. The notions of entropy and information content are discussed from the perspective of information gain.

B. Dataset for FallDS-IoT

The mentioned dataset includes 4 different outputs that could occur as a result of the fall-DS. To accomplish this, we collect the proper dataset by taking into account the various activities that a person engages in, such as sitting, sleeping, falling, walking, and then conducting experiments. Taking up these activities, which are then recorded by the sensors, is how the accumulation is carried out. We have a dataset (real-time) of 20,500 examples, each of which has a unique set of characteristics. The information obtained from each simulation is gathered to evaluate the likely behavior or the person's conditions [22]. This actual data directs important criteria in the process of learning about the activities that are most frequently performed by individuals daily. Additionally, it offers valuable statistics regarding the individual's movements and the amount of time spent in various states. For the sake of this particular job, we make use of MongoDB as the database [23].

III. Architecture of FALLDS-IOT

Additionally, the design of FallDS-IoT is presented in this section. The components that make up this system are as follows: data collection, preprocessing, classification, training, and the alert system that is utilized. Fig. 1 depicts the FallDS-IoT structure that was utilized to ensure that this research is considered comprehensive.

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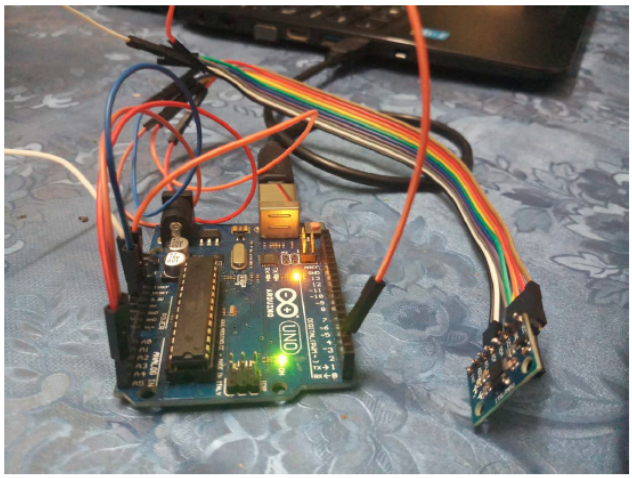


Fig. 2. A Gyroscope and Accelerometer Sensor Incorporated Into the MPU6050 Using Arduino

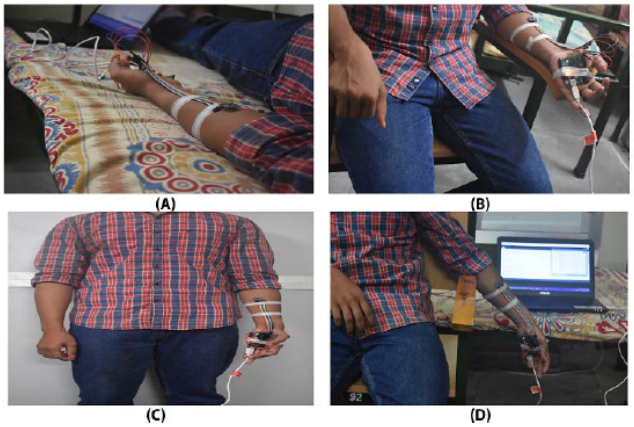


Fig. 3. Prerequisites for the Fall Detection System

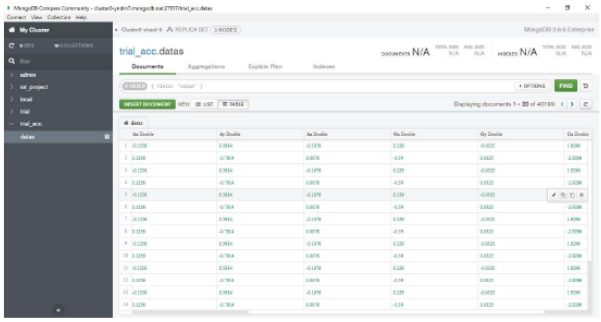


Fig. 4. MongoDB storage for the data

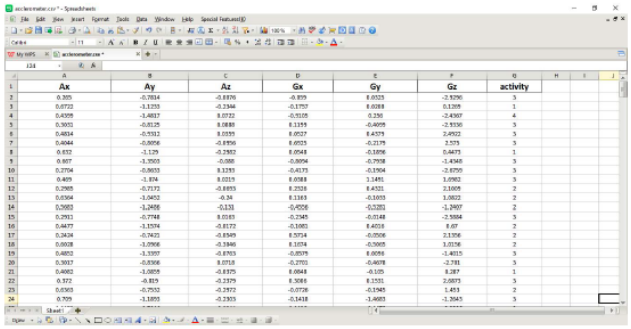


Fig. 5. Preprocessed Dataset

A. Data Gathering

The suggested system makes use of accelerometers and gyroscopes as sensors to generate and collect data on the activity level of the individual [24]. The connection between the sensors and Arduino for data gathering is depicted in Fig. 2. An accelerometer is a type of inertial dynamic sensor that can detect acceleration over a wide range of accelerations [25]. The gyroscope is a device that is used to detect angular velocity, which is a characteristic that is difficult for a human being to quantify. People's perspectives shift as a result of this phenomenon. As part of the process of generating the raw data, we obtained readings from the sensors at predetermined intervals of time. For example, the time is equivalent to five seconds. These readings are obtained concerning four different conditions, including falling, sitting, walking, and sleeping, and the values that are recorded are the ones that are saved in the database [26]. MongoDB is utilized for this purpose. Remember that MongoDB is a free and open-source database management system that is mostly utilized for document-oriented database programming. MongoDB makes use of documents that are similar to JSON and contain schemas [27]. A total of 20,000 instances were collected, which ensured that all monitoring circumstances were covered. You should be aware that the database is modulated at a predetermined interval of time. A, B, C, and D are the four circumstances that are depicted in Fig. 3. The information that was posted to MongoDB is depicted in Fig. 4.

B. Preprocessing

The investigation of the sensor data is preceded by a crucial step known as data pretreatment, which is carried out before the analysis. It is of utmost significance to identify the values that are absent from the dataset and either eliminate or substitute such values. It is possible that the classification method will not achieve the level of accuracy that is needed if it is not eliminated or replaced. An attribute name is assigned to each column of the dataset to facilitate classification of the dataset based on the data that has been recorded and the labels that have been assigned to those data. To convert the magnitude of the values obtained from the gyroscope (G) and accelerometer (A) sensors, the given equations are utilized [28].

$$A = (aX^2 + aY^2 + aZ^2)^{\frac{1}{2}} \tag{1}$$

$$G = (gX^2 + gY^2 + gZ^2)^{\frac{1}{2}} \tag{2}$$

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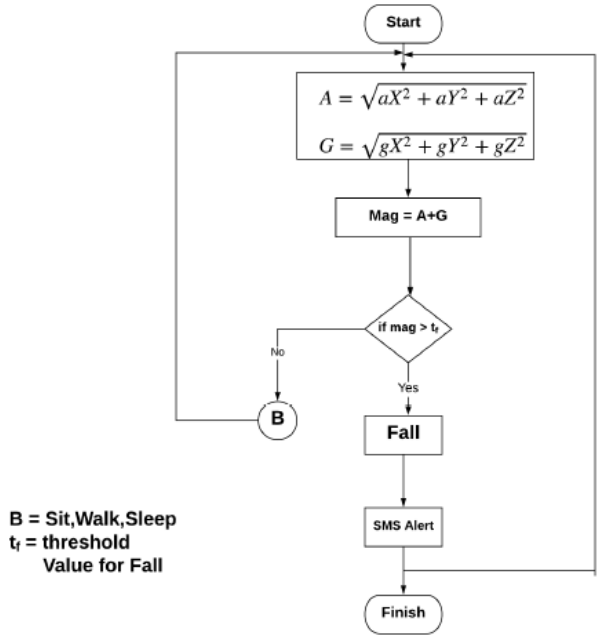


Fig. 6. Diagram of the FallDS-IoT System Flowchart

where the readings generated by the accelerometer are denoted by aX, aY, and aZ, while the readings generated by the gyroscope are denoted by gX, gY, and gZ [7]. The mentioned class labels are broken down into 4 different activities, such as sitting, sleeping, falling, and walking. The classification algorithms will make predictions about the new inputs based on these activities. In Fig. 5, the dataset that has been preprocessed is presented.

C. Training

When we are in this working step, we train the overall model by taking the dataset and using two different algorithms written in Python. A particular mapping from the input data to the target classification is performed by the learning algorithms, which are responsible for discovering the hidden patterns within the dataset. In addition to this, it generates a classification model to categorize the data that is more recent or forthcoming.

D. Classification

During this phase, the primary purpose is to utilize an ML model to train properly the data, which is subjected to training. In this case, the new data consists of the readings from the gyroscope and the accelerometer, which will be used as input for the model. An example of this would be the new observation, which will take the shape of [0.132, 0.485, 0.121, 0.526, 0.819, 0.728].

E. Alert System

An automatic call will be made by the alert system utilizing a Python program if the new observation is projected to be a fall. Additionally, a text message will be sent automatically to the mobile number that was registered, which may be attended to by a member of the family, a fall prevention center, or a medical care facility [29].

TABLE I
AN ANALYSIS OF THE K-NN'S PERFORMANCE

SL No.	Precision	Recall	F1 Score	Support
0	1.00	0.98	0.99	63
1	1.00	1.00	1.00	3449
2	1.00	1.00	1.00	7725
3	1.00	1.00	1.00	6852
4	0.99	1.00	0.99	3728
5	0.98	0.97	0.98	2178

F. Steps of FallDS-IoT

Fig. 6 depicts the actual flowchart of the FallDS-IoT implementation. These are the steps that make up the system, which are described below. Take all of the conditions into consideration and merge the sensor results at regular intervals of time. Make use of the dataset that was generated to train the model. The classification of fresh observations or inputs can be accomplished by the application of machine learning algorithms [30]. As can be seen in Fig. 6, the FallDS-IoT system is put through its paces under a variety of scenarios.

IV. Results and Discussions

To evaluate the effectiveness of the system, we make use of the Python programming language. The following parameters are utilized in the process of performing a performance analysis on categorization. It is important to keep in mind that it is the ratio of genuine positives to the sum of true positives and false negatives. The true benefits of autumn are discussed, and you may rest assured that they are identified accurately and without fail.

F1 Score: This score is used to quantify accuracy and is calculated by taking the harmonic mean of precision and recall. Support refers to the number of instances in which a variable is located within the dataset that was generated. The classification labels are then represented in the following manner. The number 0 denotes the state of being idle, the number 1 denotes standing, the number 2 denotes walking, the number 3 denotes sleeping, and the numbers 4 and 5 denote falling cases. To determine the labels, we make use of K-NN in the datasets that have been generated. Table I displays the outcome of the result. An accuracy level of 97.85% is seen to be displayed by the K-NN algorithm. The performance investigation of the decision tree is presented in Table II, which is similar to the previous one. One might observe that the decision tree demonstrates an accuracy level of 90.59 percent.

It can be seen in these two tables that K-NN is demonstrating a higher level of accuracy when it comes to identifying instances of falls. Fig. 7 depicts the K value in comparison to the curve of the error rate. Figure 8 illustrates the accelerometer and gyroscope readings that were taken.

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TABLE II
A PERFORMANCE ANALYSIS OF THE DECISION TREE

SL No.	Precision	Recall	F1 Score	Support
0	1.00	1.00	1.00	83
1	1.00	1.00	1.00	3530
2	1.00	1.00	1.00	7649
3	1.00	1.00	0.99	6805
4	0.99	0.98	0.97	3872
5	0.46	1.00	0.63	2132

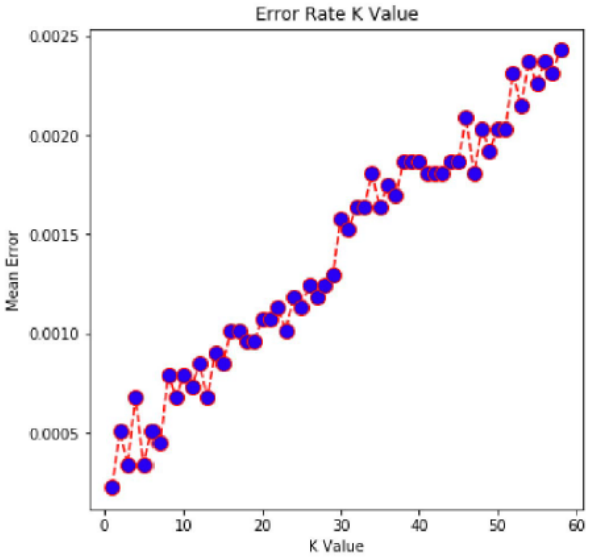


Fig. 7. In K-NN, the K value vs the Error Rate

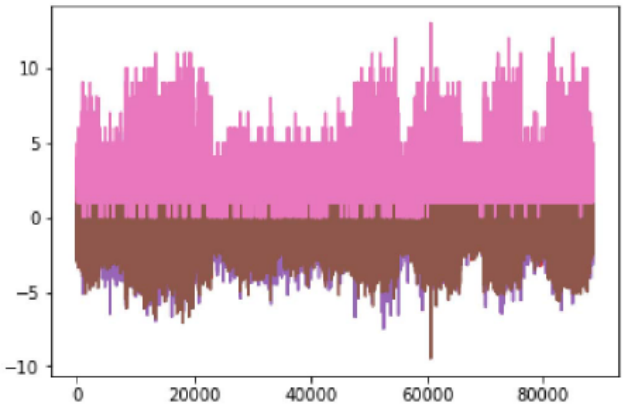


Fig. 8. Levels of the sensors' values

V. Conclusions

We have presented a Fall Detection System that is integrated with the Internet of Things (FallDS-IoT) in this particular piece of writing. It is designed to detect falls that occur among senior persons by utilizing a device that is worn on the body. To determine the outputs of the fall-DS, we have utilized gyroscope and accelerometer sensors. According to our classification system, the activities that senior persons engage in daily are as follows: sitting, sleeping, falling, and walking. Specifically, the K-NN method and the decision tree algorithm are two well-known ML techniques that we have utilized in the development of the fall-DS. The precision of these algorithms was 98.75 percent and 90.59 percent, respectively, when applied to the dataset that we recorded. Therefore, to proceed with the classification process, we have chosen K-NN. Whenever a fall occurs after classification, a rapid mobile message informing about the fall will be sent to a registered cellphone number swiftly through the Python program. This message will inform the recipient about the fall. In the future, we are going to conduct an exhaustive study of the classification methods and determine which one is the most suitable for the classification of datasets of varying sizes.

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